

2.14 Enzyme Kinetics

▼ What is the role of enzymes in our bodies?

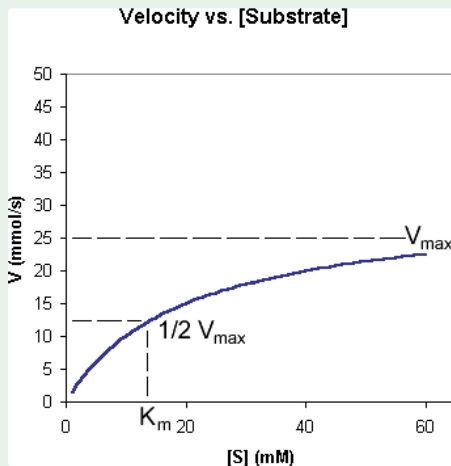
- To act as **catalysts**, ensuring that the many biochemical reactions taking place in our body **do so at a rate that can support life**.

▼ What is Michaelis constant (K_m)?

- The concentration of substrate which allows an enzyme to function at half its maximal rate
- It can be used to compare relative enzyme activity

▼ What is V_{max} ?

- **V_{max} of enzyme is the maximum rate of reaction**
 - Maximum rate of substrate processing by enzymes



▼ What is the purpose of a reaction buffer?

- The buffer **maintains the pH of the reaction mixture** and **supplies potassium (K^+) or ammonium (NH_4^+) ions for optimal enzyme activity**

- ▼ Describe the steps involved to set up this practical used to determine K_m and V_{max} of chymotrypsin.
 - Add relevant **volume of reaction buffer** to cuvette using pipette
 - Add relevant **volume of GPNA stock solution (substrate)** to cuvette using pipette
 - **Mix together** with thumb placed on top of the cuvette opening
 - Place the cuvette in the spectrophotometer, **calibrating so the absorbance reads zero**
 - Add relevant **volume of enzyme** to cuvette using pipette
 - **Place cuvette in spectrophotometer** immediately, **starting a timer** and **measuring absorbance at time intervals**
 - **Vary the concentrations of the GPNA stock solution** in order to **assess the enzyme activity at different substrate concentrations**
 - Keep the **enzyme concentration constant** at all GPNA concentrations
 - **NOTE:** As you **decrease the volume of GPNA**, you must **increase the volume of buffer solution** being added to the cuvette
- ▼ What is the general purpose of spectrophotometry?
 - To **measure the concentration** of a solution
 - **Absorbance is measured**, and **using the Beer-Lambert Law**, **concentrations** can be obtained
 - Beer Lambert Law = $A = E \times c \times \ell$ where
 - **A = absorbance**
 - **E = extinction coefficient (8.8)**
 - **c = concentration** of absorbing solution
 - **ℓ = path length (length of cuvette, not wavelength)**

▼ Describe how the Beer-Lambert Law can be used experimentally to interpret the results of this practical.

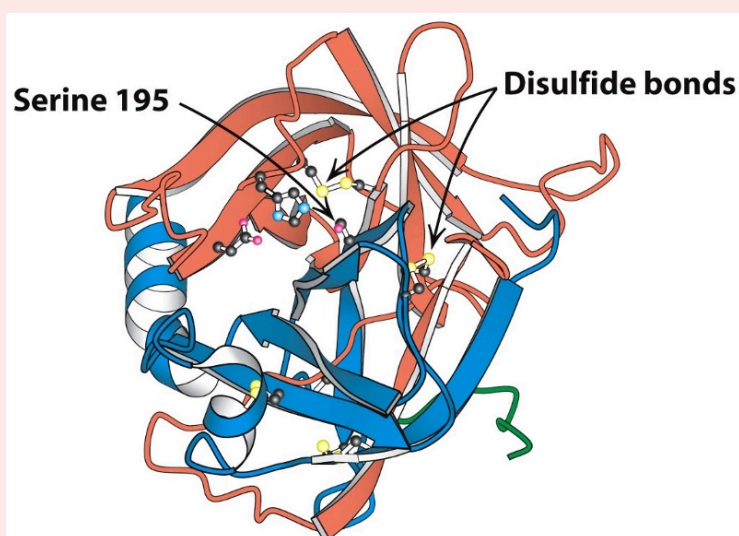
- Beer Lambert Law = $A = E \times c \times \ell$ where
 - **A = absorbance**
 - **E = extinction coefficient (8.8)**
 - **c = concentration** of absorbing solution
 - **ℓ = path length (length of cuvette, not wavelength)**
- **Concentration is the only variable** in this equation, and this **changes depending on the activity of the enzyme in solution and production of p-nitroalanine** (product). Hence, the greater the enzyme activity, the greater the concentration and hence absorbance on the spectrophotometer will read.
- **Absorbance is converted to concentration** using the Beer-Lambert Law
- A graph can be plot of concentration against time, and then, a reciprocal graph of initial rate of reaction against different substrate concentrations can be plot to obtain **K_m** and **V_{max}**

▼ What is chymotrypsin?

- **Serine protease** found in the **digestive system of mammals**.

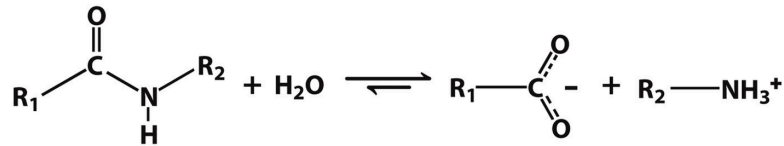
▼ Describe the structure of chymotrypsin.

- **Arranged in three peptide chains (A, B and C) linked by disulphide bridges**



- ▼ Image showing how chymotrypsin cleaves peptide bonds (hydrolysis).

Chymotrypsin cleaves certain peptide bonds



- ▼ What property on proteins does chymotrypsin show specificity for?

- Bulky hydrophobic side chains (see below benzene)

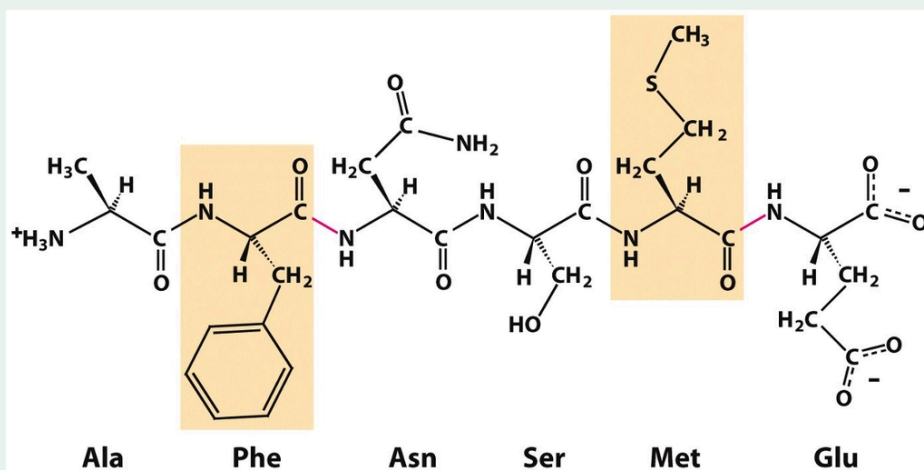
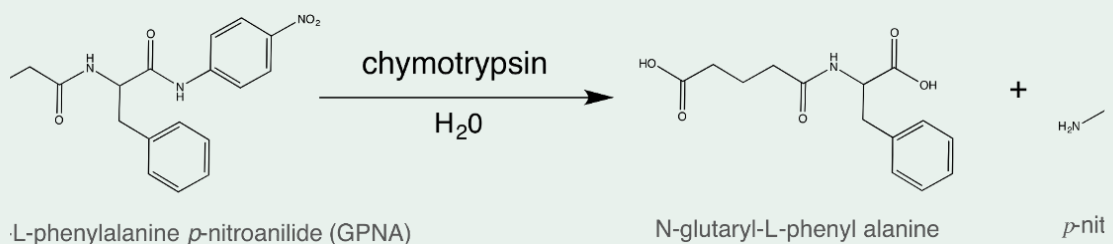


Figure 9.1
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- ▼ What reaction are we measuring in this practical?

- Breakdown of GPNA into N-glutaryl-L-phenyl alanine and p-nitroaniline
 - We are measuring p-nitroaniline in the spectrophotometer

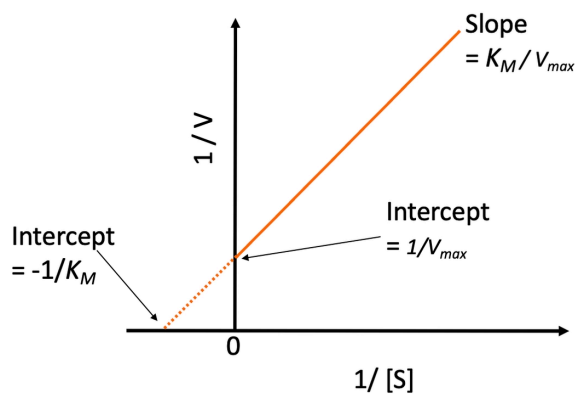


- ▼ Why are we measuring p-nitroaniline in the spectrophotometer?
 - We are measuring p-nitroaniline in the spectrophotometer **because it produces a yellow colour, and hence absorbance**
 - This **helps us to monitor product formation**
- ▼ What is V_0 ?
 - Initial rate of reaction
- ▼ What is K_m ?
 - **K_m** is known as **Michaelis Constant** and is defined as the **concentration of substrate** at which a **particular enzyme at a particular concentration works at half its maximal velocity**.
 - It helps us to compare the relative strength of enzyme-substrate complexes
- ▼ What does a high/low K_m show?
 - A **high K_m** shows typically **weak binding between enzymes and substrates** and hence **low enzymatic activity**
 - A **low K_m** shows typically **strong binding between enzymes and substrates** and hence **high enzymatic activity**
- ▼ What is a Lineweaver-Burk plot?
 - A **double reciprocal plot used to determine K_m and V_{max}**

▼ Describe how a Lineweaver-Burk plot helps us to determine K_M and V_{max} .

- The **x-intercept** = $-1/K_M$
- The **y-intercept** = $1/V_{max}$
- x-axis = $1/\text{substrate concentration}$
- y-axis = $1/\text{initial rate of reaction}$

Lineweaver-Burk plot

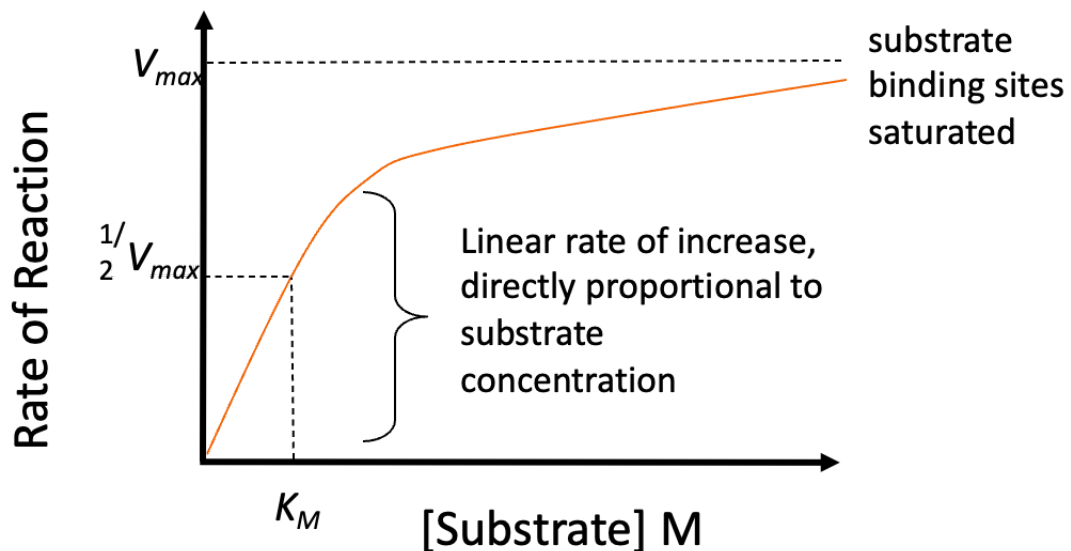


A double-reciprocal plot of $1/V$ against $1/[S]$

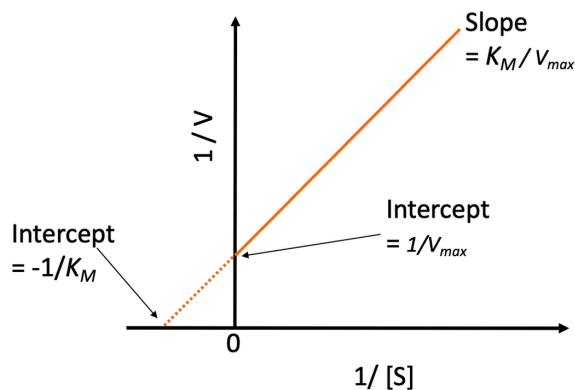
Intercepts give us K_M and V_{max} .

- ▼ Image showing how the Lineweaver-Burk plot can help us to calculate K_M and V_{max} .

- The first figure is a regular rate of reaction against substrate concentration graph.
- The second figure shows a Lineweaver-Burk plot.



Lineweaver-Burk plot



A double-reciprocal plot of $1/V$ against $1/[S]$

Intercepts give us K_M and V_{max} .

- $1/V = 1 / \text{initial rate of reaction (V for velocity)}$
- $1/[S] = 1 / \text{substrate concentration}$

- ▼ What is the turnover number (K_{cat}) of an enzyme?

- The number of bonds that a single enzyme can cleave in a second
 - i.e the number of molecules that a single enzyme can process in a given unit of time

▼ How can the turnover number (K_{cat}) of an enzyme be calculated?

- Turnover number (K_{cat}) = **V_{max} / enzyme concentration**
 - This is because V_{max} measures the **maximum rate at which product formation occurs** (and hence maximum rate at which enzyme acts on substrate) and enzyme concentration tells us **how many enzymes are involved this product formation**

▼ Which equation which relates the absorbance of a solute to its concentration?

- Beer Lambert Law = **$A = E \times c \times \ell$** where
 - **A = absorbance**
 - **E = extinction coefficient**
 - **c = concentration** of absorbing solution
 - **ℓ = path length**

▼ What effect does a competitive antagonist (one that binds in or near the enzyme's active site) have on K_m and V_{max} ?

- It **increases K_m** but has **no effect on V_{max}**
- This is because the **enzyme's function has not been affected**, but the substrate concentration required to bind to a given concentration of enzyme has increased as the **enzyme's affinity for the substrate has decreased due to increased competition for the active site** (hence the name competitive inhibitor)

▼ What effect does a non-competitive antagonist have on K_m and V_{max} ?

- **Non-competitive inhibitors have no effect on K_m** but act to **lower V_{max}**
- This is because they **induce conformational changes on the enzyme** which causes a **reduction in the V_{max}** , but **do not actually change the affinity of substrate molecules** binding to enzymes, so the same amount of substrate is required to induce similar effects

▼ Graph showing the effect that competitive antagonists have on Lineweaver-Burk plots.